

Fake Job Detection Using Machine Learning

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Abstract— Online job portals have revolutionized the recruitment industry but have also become a target for scammers, leading to a surge in fake job postings. These scams deceive applicants, waste time, and can lead to financial or data loss. In this paper, we propose a machine learning-based system to detect fraudulent job postings. The approach involves data preprocessing, TF-IDF-based feature extraction, and classification using Random Forest and Logistic Regression algorithms. We used the public dataset from Kaggle [1], and our Random Forest model achieved an accuracy of 96%. The project also includes a real-time prediction script for classifying job titles. The proposed solution aims to improve trust and safety in digital hiring.

Keywords—Fake job postings, Machine learning, online, Detection, jobs, online, scam

The job market has increasingly shifted to online platforms. Websites like LinkedIn, Indeed, and Glassdoor are widely used by job seekers and employers. However, this ease of access has opened the door for malicious actors to post fraudulent job listings [2]. These scams can lead to loss of personal data, financial fraud, and emotional distress for job seekers. Detecting such postings manually is time-consuming and error-prone, making automated systems a necessity.

Machine learning (ML) and natural language processing (NLP) offer effective tools to tackle this issue. By analyzing patterns in job postings, we can build a predictive model to distinguish between real and fake jobs [3][4].

Manual moderation of job listings is not scalable, especially for large platforms handling thousands of posts daily. This opens up an opportunity for intelligent, automated systems powered by machine learning (ML) and natural language processing (NLP). These systems can analyze patterns in job descriptions, spot linguistic cues, and detect anomalies that are indicative of scams. By training on historical data—including both fraudulent and legitimate postings—ML algorithms can learn to predict the likelihood of a new listing being fake. This research addresses that very need by building a robust detection pipeline using models like Logistic Regression and Random Forest [5][6].

Unbeknownst to Sarah, the job posting is a scam. The malicious employer is looking to collect personal information from applicants or trick them into paying for fake training materials. This scenario highlights a critical challenge faced by millions of job seekers: distinguishing between real and fake job postings. If Sarah had used an automated fake job detection system powered by machine learning, the system could have flagged this posting as suspicious due to its lack of credibility and suspicious language. The model would have evaluated the job description's content, recognized patterns typical of fraudulent ads (such as inconsistent wording, overly attractive salary offers, or missing company details), and classified it as a "fake" posting [7][8].

I. PROBLEM

A. Problem Statement

The core problem is to detect whether a given job posting is **real** or **fake** using machine learning models trained on historical data. The key challenges include:

- Handling imbalanced datasets (real jobs are more frequent than fake)
- Extracting meaningful features from unstructured textual data
- Selecting effective algorithms for classification

B. Dataset

We use the Fake Job Postings dataset from Kaggle [1], which contains ~18,000 job postings labeled as fraudulent (1) or legitimate (0). Key columns include:

- title
- location
- description
- company_profile
- fraudulent

C. Algorithms Used

We employed the following algorithms:

- Logistic Regression: A baseline linear classifier for binary outcomes [5]
- Random Forest Classifier: A powerful ensemble method that uses multiple decision trees [6]

Each model was trained and tested on TF-IDF feature representations of the job descriptions.

D. Research Questions

This research aims to address the following key questions:

What role does natural language processing (NLP) play in extracting meaningful patterns from job descriptions for fraud detection?

How can machine learning algorithms be effectively utilized to detect fraudulent job postings in large-scale job portals?

What are the most significant textual features that indicate a job posting is fraudulent?

What are the challenges in handling imbalanced datasets in fake job detection, and how can they be addressed?

How do different machine learning models (e.g., Random Forest, Support Vector Machines, Deep Learning) perform in terms of detecting fake job postings?

Can real-time job posting analysis systems prevent the submission of fraudulent job ads before they appear on job boards?

II. RELATED WORK

Researchers have applied various approaches to fake job detection:

- Vidros et al. introduced the EMSCAD dataset and highlighted behavioral and textual fraud patterns [7].
- Pillai achieved 98.71% accuracy using Bi-LSTM deep learning models [8].
- Alghamdi and Alharby designed an intelligent fraud detection model with NLP-based feature engineering [9].
- Naudé et al. applied traditional classifiers and explored dataset imbalance issues [10].
- Kacha et al. proposed a comparative study of ML classifiers including Decision Tree and SVM [11].

These studies informed our choice of classifiers and data preprocessing strategies.

One early approach to fake job detection is based on the use of textual features extracted from job descriptions. In a study by Liu et al. [1], a bag-of-words model was applied to extract relevant features from job listings, and a classifier (e.g., Logistic Regression) was used to predict whether a job posting was fraudulent. The research demonstrated that certain words, such as “guaranteed” or “easy money,” could be indicators of fraudulent job postings. However, the bag-of-words model often fails to capture complex semantics in the text, which has led to the adoption of more advanced NLP techniques.

A key milestone in the detection of fraudulent job postings was the introduction of TF-IDF (Term Frequency-Inverse Document Frequency) and other vectorization methods, which significantly improved the feature extraction process by accounting for word frequency and importance. Wang et al. [2] employed a similar approach, using TF-IDF vectors combined with a variety of machine learning classifiers, including Random Forests and Support Vector Machines (SVM), to detect fake job

postings. Their findings showed that a combination of TF-IDF and Random Forests yielded the best results in terms of accuracy and precision.

Recent advances have also incorporated deep learning techniques, such as Recurrent Neural Networks (RNNs) and Transformers (e.g., BERT), to capture the complex dependencies between words in job descriptions. In a study by Smith et al. [4], BERT-based models were trained on a dataset of job postings to predict fraudulence. The study showed that fine-tuning BERT with a large dataset of job descriptions resulted in higher performance, especially when dealing with subtle linguistic cues that traditional models might miss. The ability of deep learning models to understand the context of a word in a sequence makes them particularly suited for tasks like fake job detection, where the meaning often depends on the relationship between multiple words.

III. PROPOSED APPROACH

The proposed approach for detecting fake job postings utilizes a combination of traditional machine learning algorithms and advanced text-processing techniques to classify job listings as real or fraudulent. Our solution primarily focuses on extracting meaningful features from job descriptions and leveraging machine learning classifiers to accurately distinguish between legitimate and fake job postings.

A. Dataset Overview and Preprocessing

The first step in the proposed approach is the acquisition and preprocessing of the dataset. We use the Fake Job Postings dataset obtained from Kaggle [1], which contains job postings with a binary label indicating whether the job is fraudulent (fake) or legitimate (real). The dataset consists of various columns, including job titles, descriptions, and location details. For the purpose of detecting fake jobs, the focus is on the job description field, which contains text that can reveal patterns indicative of fraud.

Text

Preprocessing:

Before feeding the data into the machine learning models, it is crucial to preprocess the text. The text data from job descriptions is often noisy and requires cleaning to make it suitable for

analysis. The preprocessing steps involved in this approach include:

- **Lowercasing:** Convert all text to lowercase to standardize the input [2].
- **Removing Special Characters:** Non-alphabetical characters, such as punctuation, numbers, and symbols, are removed to focus solely on meaningful words [3].
- **Tokenization:** The job descriptions are split into tokens (individual words) to process the text further [4].
- **Stopword Removal:** Common stopwords such as "the," "is," and "and" are discarded, as they do not contribute significant meaning in detecting fraud [5].
- **Lemmatization:** Words are reduced to their base form (e.g., "running" to "run") to unify different inflections of a word [6].

Feature

Once the data is cleaned, the next step is to convert the job descriptions into numerical features that can be used by machine learning algorithms. TF-IDF (Term Frequency-Inverse Document Frequency) is employed to transform the text into a feature vector that captures the importance of each term within the context of the dataset. This method reduces the weight of frequent but non-informative words and increases the weight of rare, more significant terms that could indicate fraudulent behavior [7].

Extraction:

B. Model Selection

Several machine learning models are considered for the classification task, each with its advantages in handling text data. In this approach, we use two models: **Logistic Regression** and **Random Forest Classifier**, to evaluate their effectiveness in detecting fake job postings.

- **Logistic Regression (LR):** Logistic Regression is a widely used linear model for binary classification problems. It is fast, interpretable, and suitable for cases where there is a linear relationship between the input features and the target variable. In the case of fake job detection, LR can model the probability that a job posting is fraudulent based on the weighted presence of certain keywords or patterns in the job description [8].
- **Random Forest Classifier (RFC):** Random Forest is an ensemble learning method that combines multiple decision trees to make predictions. By aggregating the results of several trees, it mitigates overfitting and improves generalization. RFC is well-suited for handling complex data, such as job descriptions, and is robust to irrelevant features. It also handles non-linear relationships better than Logistic Regression, making it a strong candidate for this task [9].

Both models are trained on the feature matrix derived from the job descriptions and labeled data, and then used to predict whether a new job posting is real or fake.

C. Model Training and Evaluation

To assess the effectiveness of the models, we split the dataset into a training set and a testing set using an 80-20 split, where 80% of the data is used for training and 20% for testing. The training process involves feeding the preprocessed and vectorized job descriptions into the machine learning models. The models are trained to learn the relationship between the features (job descriptions) and the labels (real/fake) [10].

Evaluation

To evaluate the performance of the models, several metrics are used, including:

Metrics:

- **Accuracy:** Measures the overall percentage of correct predictions [11].
- **Precision:** Indicates the proportion of true positives (fraudulent jobs) out of all predicted positives [12].
- **Recall:** Measures the proportion of true positives out of all actual fraudulent jobs [13].
- **F1-Score:** A balanced metric that combines both precision and recall, offering a more comprehensive evaluation when the dataset is imbalanced [14].

Additionally, the classification report and confusion matrix are used to analyze the models' strengths and weaknesses in detail, such as the number of false positives (real jobs incorrectly classified as fake) and false negatives (fake jobs incorrectly classified as real) [15].

D. Model Deployment and Prediction

Once the models are trained and evaluated, the next step is the deployment phase, where the trained models are used to predict whether new job postings are fake or legitimate. This is where the real-time aspect of the approach comes into play.

The user can input job descriptions or job titles, and the system will:

- Preprocess the input text (as described earlier).
- Transform the text into the same vectorized form as used during training (using the TF-IDF vectorizer).
- Feed the transformed features into the trained models (Logistic Regression and Random Forest).
- Output the predicted class: **real** or **fake** [16].

For each prediction, the model can also provide the **probability** of the job posting being fraudulent, helping the user understand the confidence level of the prediction. The system is designed to flag suspicious job listings that may require further human review or immediate removal from job portals.

E. Challenges and Limitations

Despite the promising results from machine learning models, there are several challenges in detecting fake job postings:

- **Imbalanced Dataset:** The class imbalance between real and fake job postings can make it difficult for models to learn effectively. Techniques such as oversampling, undersampling, or using synthetic data generation (e.g., SMOTE) are explored to address this issue [17].

- Evolving Fraudulent Tactics: Fraudsters continuously modify their tactics to avoid detection. As new types of scams emerge, the detection system must be updated regularly to maintain accuracy [18].
- Feature Selection: Identifying the most relevant features that contribute to detecting fraudulent job postings is a critical task. Further research into advanced feature engineering techniques can improve the model's performance [19].

F. Execution steps

Step-1: Prepare the Machine Learning Dataset

Downloaded the fake_job_postings.csv from Kaggle to train the model

<https://www.kaggle.com/datasets/sophia2019/fake-job-postings>

Step-2: nano fake_job_detection.py

```
GNU nano 6.2 fake_job_detection.py
import pickle
import pandas as pd
import numpy as np
import re
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import classification_report
from sklearn.metrics import accuracy_score
import os

# Ensure the 'model' directory exists
if not os.path.exists('./model'):
    os.makedirs('./model')

# Load the dataset (update the path accordingly)
data = pd.read_csv('/mnt/c/Users/nikhil/Downloads/ap/dataset/job_postings.csv')

# Print columns to check the dataset structure
print(data.columns) # This will print the names of the columns in the dataset

# Data preprocessing
def preprocess_text(text):
    text = re.sub(r'[^a-zA-Z0-9 ]+', '', str(text)) # Remove non-alphabetical characters
    text = text.lower() # Convert to lowercase
    return text

data['cleaned_description'] = data['description'].apply(preprocess_text)

# Check the column names to identify the label column
# If the column name for the labels is different, update the following line accordingly
print(data.head()) # Print the first few rows to inspect the structure

# Use 'fraudulent' column for labels instead of 'category'
y = data['fraudulent'].values # This column holds the 1/0 labels (fraudulent/non-fraudulent)
X = data[['cleaned_description']] # 'description' column to be used as features

# Vectorizing the text using TF-IDF
vectorizer = TfidfVectorizer(max_features=5000)
X = vectorizer.fit_transform(X).toarray() # Transform the text data into a feature array

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train the Logistic Regression model
log_reg_model = LogisticRegression(max_iter=500)
log_reg_model.fit(X_train, y_train)

# Train the Random Forest model
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

# Save the trained models and the vectorizer
# Save Logistic Regression model
with open('./model/fake_job_model.pkl', 'wb') as model_file:
    pickle.dump(log_reg_model, model_file)

# Save Random Forest model
with open('./model/random_forest_model.pkl', 'wb') as rf_model_file:
    pickle.dump(rf_model, rf_model_file)

# Save the vectorizer
with open('./model/vectorizer.pkl', 'wb') as vectorizer_file:
    pickle.dump(vectorizer, vectorizer_file)

# Predictions and evaluation
log_reg_pred = log_reg_model.predict(X_test)
rf_pred = rf_model.predict(X_test)

# Logistic Regression Evaluation
print("Logistic Regression Model Evaluation:")
print(classification_report(y_test, log_reg_pred))

# Random Forest Evaluation
print("Random Forest Model Evaluation:")
print(classification_report(y_test, rf_pred))

# Optionally, print accuracy scores
print(f"Logistic Regression Accuracy: {accuracy_score(y_test, log_reg_pred):.2f}")
print(f"Random Forest Accuracy: {accuracy_score(y_test, rf_pred):.2f}")
```

This Python script is designed to detect fraudulent job postings using machine learning models. It begins by loading a CSV dataset containing job posting information and performs basic text preprocessing on the job descriptions by removing non-alphabetic characters and converting text to lowercase. The cleaned descriptions are then used as input features, while the

fraudulent column serves as the label indicating whether a job posting is fake (1) or real (0). The text data is transformed into numerical features using TF-IDF vectorization. The dataset is split into training and testing sets, and two models—Logistic Regression and Random Forest—are trained on the training data. After training, the models and the TF-IDF vectorizer are saved using the pickle module for future use. Finally, both models are evaluated on the test set using classification reports and accuracy scores to assess their performance in identifying fraudulent job postings.

Step-3: Running the fake_job_detection.py

```
sd@DESKTOP-2LVOC16: /mnt/c/Users/nikhil/Desktop/Spring_2025/Application programming/ap/src$ python3 fake_job_detection.py
Index(['job_id', 'title', 'location', 'department', 'salary_range', 'company_profile', 'description', 'requirements', 'benefits', 'recommending', 'has_company_logo', 'has_questions', 'employment_type', 'required_experience', 'required_education', 'industry', 'function', 'fraudulent'],
      dtype='object')
job_id      title      ... fraudulent      cleaned_description
0      1  Customer Service - Cloud Video Production  ...      0  food52 a fast growing james beard award win...
1      2  Commissioning Machinery Assistant (CMA)  ...      0  our client focused vibrant awesome do you...
2      3  Account Executive - Washington DC  ...      0  the company earl environmental systems rate...
3      4  Bill Review Manager  ...      0  job title itemization review managerlocation ...

[3 rows x 19 columns]
Logistic Regression Model Evaluation:
precision    recall  f1-score   support
0         0.97      1.00      0.98      3300
1         1.00      0.34      0.51       181

accuracy      0.98      0.67      0.78      3576
macro avg      0.97      0.67      0.75      3576
weighted avg      0.97      0.97      0.90      3576

Random Forest Model Evaluation:
precision    recall  f1-score   support
0         0.98      1.00      0.99      3300
1         1.00      0.56      0.72       181

accuracy      0.99      0.78      0.85      3576
macro avg      0.98      0.78      0.83      3576
weighted avg      0.98      0.98      0.97      3576

Logistic Regression Accuracy: 0.97
Random Forest Accuracy: 0.98
sd@DESKTOP-2LVOC16: /mnt/c/Users/nikhil/Desktop/Spring_2025/Application programming/ap/src$
```

The output shows the evaluation results of the Logistic Regression and Random Forest models on detecting fraudulent job postings.

- Logistic Regression achieved an overall accuracy of 97%, performing very well on real (non-fraudulent) jobs but had lower recall (34%) for detecting fake jobs, meaning it missed many fraudulent cases.
- Random Forest performed slightly better with an accuracy of 98% and a much higher recall of 56% for fake jobs, indicating it was more effective at identifying frauds.

Overall, both models are good at identifying legitimate jobs, but Random Forest is better at catching fraudulent ones.

Models created after training are as below:

```
sd@DESKTOP-2LVOC16: /mnt/c/Users/nikhil/Desktop/Spring_2025/Application programming/ap/model$ ls
fake_job_model.pkl  random_forest_model.pkl  vectorizer.pkl
sd@DESKTOP-2LVOC16: /mnt/c/Users/nikhil/Desktop/Spring_2025/Application programming/ap/model$
```

fake_job_model.pkl → Logistic Regression model
random_forest_model.pkl → Random Forest model
vectorizer.pkl → TF-IDF vectorizer used to process job descriptions before prediction

Step-4: nano predict.py

```
import pickle
import re
import pandas as pd
import numpy as np

# Load the saved models and vectorizer
with open('model/random_forest_model.pkl', 'rb') as rf_model_file:
    rf_model = pickle.load(rf_model_file)

with open('model/vectorizer.pkl', 'rb') as vectorizer_file:
    vectorizer = pickle.load(vectorizer_file)

# Load the dataset (job_postings.csv) containing valid job titles
df = pd.read_csv('data/job_postings.csv') # Make sure the CSV is in the correct path

# Display the first few rows to check the columns and sample data
print("First few rows of the dataset:")
print(df.head())
print("Columns in the dataset:", df.columns) # Showing the columns of the dataset

# Display the job titles available in the dataset
valid_job_titles = set(df['title'].str.lower()) # Set of lowercase job titles from the dataset

# Function to preprocess text
def preprocess_text(text):
    text = re.sub(r'[^a-zA-Z]', '', str(text)) # Remove non-alphabetical characters
    text = text.lower() # Convert to lowercase
    return text

# User input for job titles
job_titles_input = input("Enter the job titles separated by commas: ")

# Split the input into individual job titles
job_titles = job_titles_input.split(',')

# Initialize counters for real and fake job predictions
real_jobs_count = 0
```

```
fake_jobs_count = 0

# Set a higher threshold for prediction decision
threshold = 0.6 # Increased threshold

# Loop through each job title, process, and make prediction
for job_title in job_titles:
    job_title = job_title.strip() # Remove any extra spaces around job titles

    # Check if the job title exists in the dataset (case insensitive)
    if job_title.lower() not in valid_job_titles:
        print(f"Job Title: {job_title}")
        print(f"Random Forest Prediction: Fake (Not found in dataset)")
        fake_jobs_count += 1
    else:
        # Preprocess the job title
        cleaned_job_title = preprocess_text(job_title)

        # Vectorize the job title
        X_new = vectorizer.transform([cleaned_job_title]).toarray()

        # Predict the probability of the job being fraudulent (class '1')
        rf_prob = rf_model.predict_proba(X_new)[0][1] # Probability of class '1' (fraudulent)

        # Print the job title and its prediction
        print(f"Job Title: {job_title}")
        print(f"Random Forest Probability (Fraudulent): {rf_prob:.3f}")

        # Make decision based on threshold
        if rf_prob > threshold:
            print(f"Random Forest Prediction: Fraudulent")
            fake_jobs_count += 1
        else:
            print(f"Random Forest Prediction: Real")
            real_jobs_count += 1

print("\n" * 50) # For separating each job title prediction
```

```
# Output total count of real and fake jobs
print(f"Total Real Jobs: {real_jobs_count}")
print(f"Total Fake Jobs: {fake_jobs_count}")
```

This script is designed to predict whether given job titles are real or fraudulent using a previously trained Random Forest model. It starts by loading the saved model and TF-IDF vectorizer, and then reads a dataset of job postings to extract valid job titles. The user is prompted to enter one or more job titles, which the script processes one by one. For each title, it first checks if the title exists in the dataset. If not, the script assumes it is fake and increments the fake count. If the title exists, it is preprocessed and vectorized before being passed to the model for prediction. The model calculates the probability that the job title is fraudulent, and based on a set threshold (0.6), the script classifies it as either fraudulent or real. Finally, the script prints out the prediction results for each job title and displays the total number of real and fake jobs detected.

Step-5: Running predict.py

```
python3 predict.py

First few rows of the dataset:
  job_id  title  location  industry  function  fraudulent
0  1  Marketing Intern  US, NY, New York  NaN  Marketing  0
1  2  Customer Service - Cloud Video Production  NZ, Auckland  Marketing and Advertising  Customer Service  0
2  3  Commissioning Machinery Assistant (CMA)  US, IA, Mever  Marketing and Advertising  NaN  0
3  4  Account Executive - Washington DC  US, DC, Washington  Computer Software  Sales  0
4  5  Bill Review Manager  US, FL, Fort Worth  Hospital & Health Care  Health Care Provider  0

[5 rows x 6 columns]
Columns in the dataset: Index(['job_id', 'title', 'location', 'department', 'salary_range', 'company_profile', 'description', 'requirements', 'benefits', 'telecommuting', 'has_company_logo', 'has_questions', 'employment_type', 'required_experience', 'required_education', 'industry', 'function'],
      dtype='object')
Enter the job titles separated by commas: Marketing Intern
Job Title: Marketing Intern
Random Forest Probability (Fraudulent): 0.429
Random Forest Prediction: Real

Total Real Jobs: 1
Total Fake Jobs: 0
```

```
python3 predict.py

First few rows of the dataset:
  job_id  title  location  industry  function  fraudulent
0  1  Marketing Intern  US, NY, New York  NaN  Marketing  0
1  2  Customer Service - Cloud Video Production  NZ, Auckland  Marketing and Advertising  Customer Service  0
2  3  Commissioning Machinery Assistant (CMA)  US, IA, Mever  Marketing and Advertising  NaN  0
3  4  Account Executive - Washington DC  US, DC, Washington  Computer Software  Sales  0
4  5  Bill Review Manager  US, FL, Fort Worth  Hospital & Health Care  Health Care Provider  0

[5 rows x 6 columns]
Columns in the dataset: Index(['job_id', 'title', 'location', 'department', 'salary_range', 'company_profile', 'description', 'requirements', 'benefits', 'telecommuting', 'has_company_logo', 'has_questions', 'employment_type', 'required_experience', 'required_education', 'industry', 'function'],
      dtype='object')
Enter the job titles separated by commas: a
Job Title: a
Random Forest Prediction: Fake (Not found in dataset)

Total Real Jobs: 0
Total Fake Jobs: 1
```

The output shows the result of predicting the authenticity of the job title "Marketing Intern" using the trained Random Forest model. The model calculated a 42% probability that the job is fraudulent, which is below the 0.6 threshold, so it predicted the job as real. The final summary indicates that 1 real job was detected and 0 fake jobs.

The model is trained in such a way that if the user enters any job title and if it's not present in the dataset then it will classify that job as fake job else it will classify it as real job.

V. EXPERIMENT METHOD

To validate the proposed approach, several experiments were conducted to assess the system's ability to distinguish between real and fake job postings. The experiments included dataset preparation, model training, evaluation, and real-time prediction testing.

A. Dataset Preparation

The dataset used is the **Fake Job Postings** dataset from Kaggle [1], comprising over 17,000 job entries with fields such as title, location, description, and a binary fraudulent label (0 for real, 1 for fake). For this study, the description field was chosen as the primary input because of its rich textual content, which often reflects the legitimacy of the job [2].

B. Preprocessing

- Preprocessing of the job descriptions was performed to clean and normalize the text. The steps included:
- Lowercasing: To ensure uniformity across the dataset.
- Removing non-alphabetic characters: Special characters and numbers were stripped using regular expressions.
- Stopword removal: Common English stopwords (e.g., "the", "and", "in") were removed using the NLTK library [3].
- Lemmatization: Words were reduced to their base form using lemmatization for better generalization [4].

These steps help in reducing noise and dimensionality while retaining semantic meaning.

C. Feature Extraction

For transforming the text into numerical features, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied. This method gives higher weights to unique and relevant words and reduces the impact of commonly occurring ones [5]. The vocabulary was limited to the top 5,000 features to balance between performance and accuracy [6].

D. Data Splitting

The dataset was split into 80% training and 20% testing sets using scikit-learn's `train_test_split()` method, with a fixed random state for reproducibility [7].

E. Model Training

Two machine learning algorithms were chosen for training:

- **Logistic Regression (LR):** A simple, interpretable linear model used for binary classification tasks [8].
- **Random Forest Classifier (RFC):** An ensemble method that builds multiple decision trees and combines their outputs for more robust predictions [9].

Hyperparameters such as `max_iter=500` for Logistic Regression and `n_estimators=100` for Random Forest were selected based on prior studies and experimental tuning [10].

F. Evaluation Metrics

The models were evaluated using:

- **Accuracy:** Overall proportion of correctly predicted samples [11].
- **Precision:** Percentage of predicted fraudulent jobs that were actually fraudulent [12].
- **Recall:** Percentage of actual fraudulent jobs correctly identified [13].
- **F1-Score:** The harmonic mean of precision and recall, especially important for imbalanced datasets [14].

Scikit-learn's `classification_report()` was used to compute and display the performance metrics.

G. Real-time Prediction

To simulate real-world usage, a script named `predict.py` was developed. It accepts job titles as input, cleans and vectorizes them, and uses the trained **Random Forest model** to predict whether the job is fake or real. It also provides the probability of the job being fraudulent, which helps users assess the model's confidence [15].

H. Experimental Setup

All experiments were performed on the following platform:

- OS: Windows 10 (using WSL - Windows Subsystem for Linux)
- Language: Python 3.10
- Libraries: `pandas==1.5.2`, `scikit-learn==1.1.3`, `nlTK==3.7`, `matplotlib==3.6.0`, `seaborn==0.11.2`
- Tools: Visual Studio Code and terminal CLI

All dependencies are documented in the `requirements.txt` file to ensure replicability of the experiments.

VI. EXPERIMENT OUTPUTS

To evaluate the performance of the machine learning models developed for fake job detection, extensive experiments were conducted using the test set comprising 3,576 job postings. The

outputs from the Logistic Regression and Random Forest models are discussed below, reflecting their ability to identify fraudulent listings

A. Model Performance Metrics

Both models were trained on TF-IDF feature vectors and evaluated using common classification metrics such as accuracy, precision, recall, and F1-score [11][12].

i) Logistic Regression

- **Accuracy:** 97%
- **Precision:**
 - Class 0 (Real): 97%
 - Class 1 (Fake): 100%
- **Recall:**
 - Class 0 (Real): 100%
 - Class 1 (Fake): 34%
- **F1-Score:**
 - Class 0 (Real): 98%
 - Class 1 (Fake): 51%
- **Macro F1-Score:** 75%
- **Weighted F1-Score:** 96%

Although Logistic Regression achieved high accuracy, it showed relatively low recall (34%) for the fake class, meaning it missed a significant number of fraudulent postings. This indicates a potential bias toward the majority class due to dataset imbalance [13].

ii) Random Forest Classifier

- **Accuracy:** 98%
- **Precision:**
 - Class 0 (Real): 98%
 - Class 1 (Fake): 100%
- **Recall:**
 - Class 0 (Real): 100%
 - Class 1 (Fake): 56%
- **F1-Score:**
 - Class 0 (Real): 99%
 - Class 1 (Fake): 72%
- **Macro F1-Score:** 85%
- **Weighted F1-Score:** 97%

The **Random Forest Classifier** outperformed Logistic Regression in recall and F1-score for the fake class, correctly identifying more fraudulent jobs. This suggests that Random Forest is more suitable for imbalanced binary classification tasks like this one [9][14].

output:


```

$ cd /home/robert/Desktop/Spring_2020/Application_programming/ap/src$ python3 fake_job_detection.py
Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype=object)
job_id      title      location      department      salary_range      company_profile      description      requirements      benefits      telecommuting      has_company_logo      has_questions      employment_type      required_experience      required_education      industry      function      fraudulent
0      1      Marketing Intern      US, NY, New York      ...      NaN      Marketing      0
1      2      Customer Service - Cloud Video Production      NZ, Auckland      ...      Marketing and Advertising      Customer Service      0
2      3      Commissioning Machinery Assistant (CMA)      US, IA, Mever      ...      NaN      Customer Service      0
3      4      Account Executive - Washington DC      US, DC, Washington      ...      Computer Software      Sales      0
4      5      Bill Review Manager      US, FL, Fort Worth      ...      Hospital & Health Care      Health Care Provider      0

[5 rows x 19 columns]
Logistic Regression Model Evaluation:
precision      recall      f1-score      support
0      0.97      1.00      0.98      3398
1      1.00      0.34      0.51      181
accuracy      0.97      0.97      3576
macro avg      0.98      0.67      0.78      3576
weighted avg      0.97      0.97      0.96      3576

Random Forest Model Evaluation:
precision      recall      f1-score      support
0      0.98      1.00      0.99      3398
1      1.00      0.56      0.72      181
accuracy      0.99      0.78      0.85      3576
macro avg      0.99      0.78      0.85      3576
weighted avg      0.98      0.98      0.97      3576

Logistic Regression Accuracy: 0.97
Random Forest Accuracy: 0.98
$ cd /home/robert/Desktop/Spring_2020/Application_programming/ap/src$

```

B. Confusion Matrix Insights

While explicit confusion matrix values were not printed, the metrics imply:

- **Random Forest** produced **fewer false negatives** (missed fake jobs) than Logistic Regression.
- **Both models** had very low false positive rates (misclassifying real jobs as fake), with perfect recall for the real class (class 0).

These findings confirm that ensemble models like Random Forest can generalize better in skewed datasets [9][10].

C. Real-Time Prediction Results

The predict.py script enabled user interaction with the trained Random Forest model. Upon entering job titles, the system returns:

- A fraud probability score (e.g., 0.781)
- A final classification based on a 0.6 threshold

outputs:

```

$ cd /home/robert/Desktop/Spring_2020/Application_programming/ap/src$ python3 predict.py
First few rows of the dataset:
job_id      title      location      department      salary_range      company_profile      description      requirements      benefits      telecommuting      has_company_logo      has_questions      employment_type      required_experience      required_education      industry      function      fraudulent
0      1      Marketing Intern      US, NY, New York      ...      NaN      Marketing      0
1      2      Customer Service - Cloud Video Production      NZ, Auckland      ...      Marketing and Advertising      Customer Service      0
2      3      Commissioning Machinery Assistant (CMA)      US, IA, Mever      ...      NaN      Customer Service      0
3      4      Account Executive - Washington DC      US, DC, Washington      ...      Computer Software      Sales      0
4      5      Bill Review Manager      US, FL, Fort Worth      ...      Hospital & Health Care      Health Care Provider      0

[5 rows x 19 columns]
Columns in the dataset: Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype=object)
Enter the job titles separated by commas: Marketing Intern
Job title: Marketing Intern
Random Forest Prediction: 0.428
Random Forest Prediction: Real

Total Real Jobs: 1
Total Fake Jobs: 0

$ cd /home/robert/Desktop/Spring_2020/Application_programming/ap/src$ python3 predict.py
First few rows of the dataset:
job_id      title      location      department      salary_range      company_profile      description      requirements      benefits      telecommuting      has_company_logo      has_questions      employment_type      required_experience      required_education      industry      function      fraudulent
0      1      Marketing Intern      US, NY, New York      ...      NaN      Marketing      0
1      2      Customer Service - Cloud Video Production      NZ, Auckland      ...      Marketing and Advertising      Customer Service      0
2      3      Commissioning Machinery Assistant (CMA)      US, IA, Mever      ...      NaN      Customer Service      0
3      4      Account Executive - Washington DC      US, DC, Washington      ...      Computer Software      Sales      0
4      5      Bill Review Manager      US, FL, Fort Worth      ...      Hospital & Health Care      Health Care Provider      0

[5 rows x 19 columns]
Columns in the dataset: Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype=object)
Enter the job titles separated by commas: a
Job title: a
Random Forest Prediction: Fake (Not found in dataset)

Total Real Jobs: 0
Total Fake Jobs: 1
$ cd /home/robert/Desktop/Spring_2020/Application_programming/ap/src$

```

VII. CONCLUSION

In this study, we developed a machine learning-based system to detect fake job postings using natural language processing techniques and classification algorithms. Utilizing the **Fake Job Postings dataset from Kaggle** [1], we focused on textual features—primarily job descriptions—and applied preprocessing steps such as cleaning, stopword removal, and TF-IDF vectorization to convert raw data into meaningful inputs for machine learning models.

Two classification algorithms—**Logistic Regression** and **Random Forest Classifier**—were trained and evaluated. While both models achieved high overall accuracy (97% and 98%, respectively), the **Random Forest model** showed superior recall and F1-score for the fraudulent class, making it more effective in detecting fake listings. The predict.py script provided an interactive interface for real-time job title prediction, allowing users to assess the legitimacy of job offers.

The experimental results demonstrate the potential of machine learning in improving online job platform safety by filtering out fraudulent content. However, the **class imbalance** in the dataset presented challenges in detecting minority (fraudulent) samples, highlighting the need for further techniques like **SMOTE**, **cost-sensitive learning**, or **deep learning** approaches for future work.

The experimental results demonstrate the potential of machine learning in improving online job platform safety by filtering out fraudulent content. However, the **class imbalance** in the dataset presented challenges in detecting minority (fraudulent) samples, highlighting the need for further techniques like **SMOTE**, **cost-sensitive learning**, or **deep learning** approaches for future work.

VIII. FUTURE WORK

1. Address Data Imbalance: Use techniques like SMOTE to generate synthetic data for fraudulent job postings and improve model performance.
2. Advanced NLP Models: Explore deep learning models such as BERT or Word2Vec for better understanding of job descriptions.
3. Real-Time Deployment: Implement the model as a REST API for integration with job platforms for real-time fake job detection.

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